Rate of changes

a	b	x	**Average rate** is $\frac{f(b)-f(a)}{b-a}=b+a$	**Instantaneous rate** is $f'(x)=2x$
0	0.5	0.5	**0.5**	**1**
0.5	1	1	**1.5**	**2**
1	1.5	1.5	**2.5**	**3**
1.5	2	2	**3.5**	**4**

Table 7.1

7.2.2 Derivative as rate of change

We have seen how the derivative is used to determine slope. The derivative can also be used to determine the rate of change of one variable with respect to another. A few examples are population growth rates, production rates, water flow rates, velocity, and acceleration.

A common use of rate of change is to describe the motion of an object moving in a straight line. In such problems, it is customary to use either a horizontal or a vertical line with a designated origin to represent the line of motion. On such lines, movements in the forward direction considered to be in the positive direction and movements in the backward direction is considered to be in the negative direction.

The function $s(t)$ that gives the position (relative to the origin) of an object as a function of time t is called a position function. It is denoted by $s=f(t)$. The velocity and the acceleration at time t is denoted as $v(t)=\frac{ds}{dt}$, and $a(t)=\frac{dv}{dt}=\frac{d^2s}{dt^2}$.

Remark

The following remarks are easy to observe:

(1) Speed is the absolute value of velocity regardless of direction and hence,

Speed $=|v(t)|=\left|\frac{ds}{dt}\right|$.

(2)
- When the particle is at rest then $v(t)=0$.
- When the particle is moving forward then $v(t)>0$.
- When the particle is moving backward then $v(t)<0$.
- When the particle changes direction, $v(t)$ then changes its sign.

(3) If t_c is the time point between the time points t_1 and t_2 $(t_1<t_c<t_2)$ where the particle changes direction then the total distance travelled from time t_1 to time t_2 is calculated as

$|s(t_1)-s(t_c)|+|s(t_c)-s(t_2)|$.

(4) Near the surface of the planet Earth, all bodies fall with the same constant acceleration. When air resistance is absent or insignificant and only force acting on a falling body is the force of gravity, we call the way the body falls is a free fall.

Chapter 7 Differential Calculus Original-new.indd 3 26-02-2020 17:21:07